GLOBAL SEISMIC TRENDS

Data-Driven Earthquake Insights

Project Report | USGS earthquake data | October 2026

Prepared as project documentation for the GUVI AI/ML mini project.

1. Executive Summary

This project builds a data-driven workflow for exploring global earthquake records from the United States Geological Survey (USGS). The workflow retrieves and preprocesses earthquake data, stores the cleaned records in MySQL, analyzes patterns using SQL and Python, and presents interactive visualizations through a Streamlit dashboard.

The processed dataset contains 739,785 records with unique earthquake IDs. The dashboard's overview reports 739,730 records with non-null magnitude, depth, and year values. The observed time range is 24 October 2021 to 24 October 2026 (UTC), with the exact endpoints recorded during data validation.

2. Problem Statement and Objectives

Earthquake catalogues contain many records and attributes. A structured analysis workflow helps users examine event strength, depth, timing, reporting networks, event classifications, tsunami flags, and geographic distribution.

- Retrieve a five-year window of earthquake records from the USGS data service.
- Clean and standardize timestamps, numeric fields, and text fields; derive time and depth categories.
- Store the cleaned dataset in a MySQL database and validate record counts and ID uniqueness.
- Use SQL and Python to answer analytical questions about earthquake patterns and data quality.
- Build an interactive Streamlit dashboard with filters, summary metrics, charts, a world map, and CSV export.
- Document findings, assumptions, and limitations clearly.

3. Data Source and Dataset

Source: USGS earthquake catalogue/API. The dataset covers a rolling five-year window from 2021-10-24 through 2026-10-24 UTC, as represented by the retrieved records.

Key validation results:

- Total processed records: 739,785.
- Unique earthquake IDs: 739,785; no duplicate IDs were found in the validation query.
- Records with non-null magnitude, depth, and year: 739,730.
- Difference: 55 records do not have all three of those fields populated.
- The 2021 and 2026 calendar years are partial years in this collection; comparisons against complete years require caution.

The table includes fields such as id, time, updated, latitude, longitude, depth_km, mag, magType, place, status, tsunami, sig, net, nst, rms, gap, magError, depthError, magNst, locationSource, magSource, types, ids, sources, type, year, month, day, day_of_week, location_region, depth_category, strong_earthquake, and major_earthquake.

4. Technology Stack

- Python: data retrieval, preprocessing, exploratory analysis, and geographic matching.

- Pandas and NumPy: tabular data manipulation and numerical processing.
- GeoPandas: spatial matching of earthquake coordinates to country polygons for selected country/continent analyses.
- MySQL 8.4: persistent storage and SQL-based analytical queries.
- mysql-connector-python: Python-to-MySQL connectivity.
- Streamlit: interactive dashboard application.
- Plotly: interactive charts and geographic visualization.
- Jupyter Notebook: development and analysis workflow.

5. Data Preparation and Validation

The preparation workflow standardizes event timestamps, converts relevant fields to numeric types, normalizes text fields, and derives calendar attributes such as year, month, day, and day of week. Depth categories and earthquake-strength flags are also derived for analysis.

The cleaned data was loaded into the MySQL database global_seismic_trends, in the earthquakes table. A count and distinct-ID validation returned 739,785 rows and 739,785 distinct IDs.

For selected geographic summaries, coordinates were matched against Natural Earth country polygons. 559,773 events were matched to a country and 180,012 were not matched. Unmatched records are likely to include ocean events and points outside or on polygon boundaries. Country/continent summaries therefore represent mapped records only and may be biased toward land-associated events.

6. MySQL Analysis

SQL queries were used to rank strong and deep earthquakes, count events by time periods, summarize magnitude types, compare reviewed and automatic status, group events by type and reporting network, analyze tsunami flags, and calculate year-over-year changes. SQL was also used for data-quality checks involving station coverage and geographic summaries where suitable mapped data was available.

Selected findings

- **Strongest event:** The highest-magnitude record in the dataset is the 2025 Kamchatka Peninsula, Russia event, magnitude 8.8, at 35 km depth.
- **Deepest event:** The maximum recorded depth in the dataset is approximately 683.6 km, in the Vanuatu region.
- **Year counts:** 2024 has the highest complete-calendar-year count in the SQL summary, with 155,899 records. 2021 and 2026 are partial years.
- **Month:** July has the highest monthly count in the summary, with 67,170 records.
- **Day of week:** Thursday has the highest count (106,951); Saturday has the lowest (103,760). The difference is modest.
- **Reporting network:** Network code ak has the most records (177,240). This reflects reporting/catalogue coverage, not necessarily the true spatial frequency of earthquakes.
- **Review status:** 658,081 records are marked reviewed and 81,704 automatic.
- **Event type:** 722,517 records are classified as earthquake; other event types include quarry blasts, explosions, and ice quakes.

- **Tsunami flag:** The annual count of records with tsunami=1 peaks at 142 in 2025. The flag should not be interpreted as proof that a damaging tsunami occurred.
- **Year-over-year counts:** 2024 has the largest complete-year count. Growth comparisons involving partial years should not be interpreted as like-for-like annual growth.

7. Python and Geographic Analysis

Python was used for exploratory analysis and selected calculations that are awkward to express in a single SQL query. Geographic analysis joined earthquake coordinates to country polygons, then summarized depth, magnitude, and event frequency for mapped records. Country-level thresholds (for example, minimum record counts) and the composite regional activity score are analysis choices made for this project, not official seismological standards.

For proximity analysis, a spatial index based on the haversine distance was used to generate nearby candidate events, followed by a time-window check. The reported output contained 1,011,290 candidate relationships passing the implemented distance/time validation. This should be described as pairs under the chosen algorithm and rules; it is not a count of distinct earthquake sequences or causal relationships.

8. Streamlit Dashboard

The dashboard connects to MySQL and allows users to explore the dataset interactively. The sidebar provides year, magnitude, and depth filters. The main page displays overview metrics and multiple charts, a global earthquake map, a strongest-event table, and a filtered-data CSV download.

- Overview metrics: filtered record count, strongest magnitude, average magnitude, and maximum depth.
- Yearly trend line: earthquake record counts by calendar year.
- Magnitude histogram: distribution of recorded magnitude values.
- Depth histogram: distribution of earthquake depths.
- Reporting-network chart: most represented network codes by record count.
- Event-type chart: counts by event classification.
- World map: geographic locations of a sample of up to 5,000 filtered records to keep rendering responsive.
- Strongest-event table: ranked records with time, place, magnitude, depth, tsunami flag, and status.
- CSV export: downloads the currently filtered earthquake records.

9. Dashboard Screenshots

The following screenshots are embedded from the running local Streamlit dashboard. Each image is sized to fit the report page.

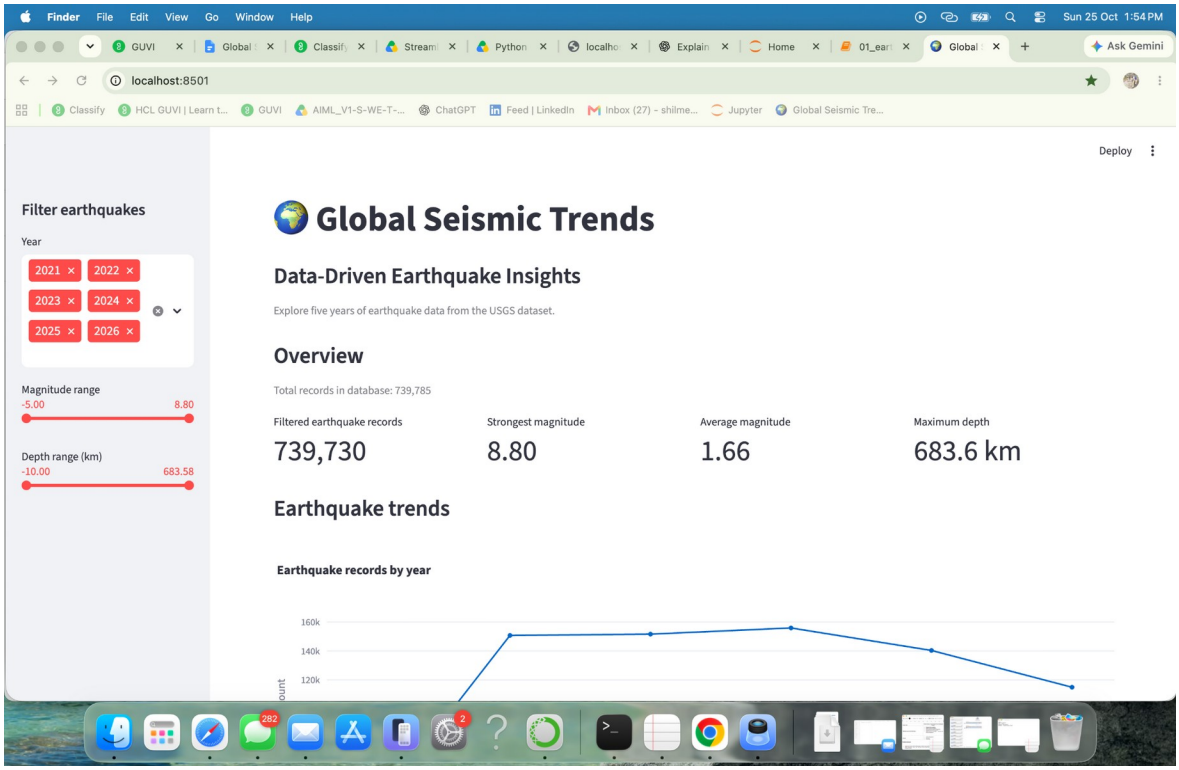


Figure 1. Dashboard overview with summary metrics and yearly trend.

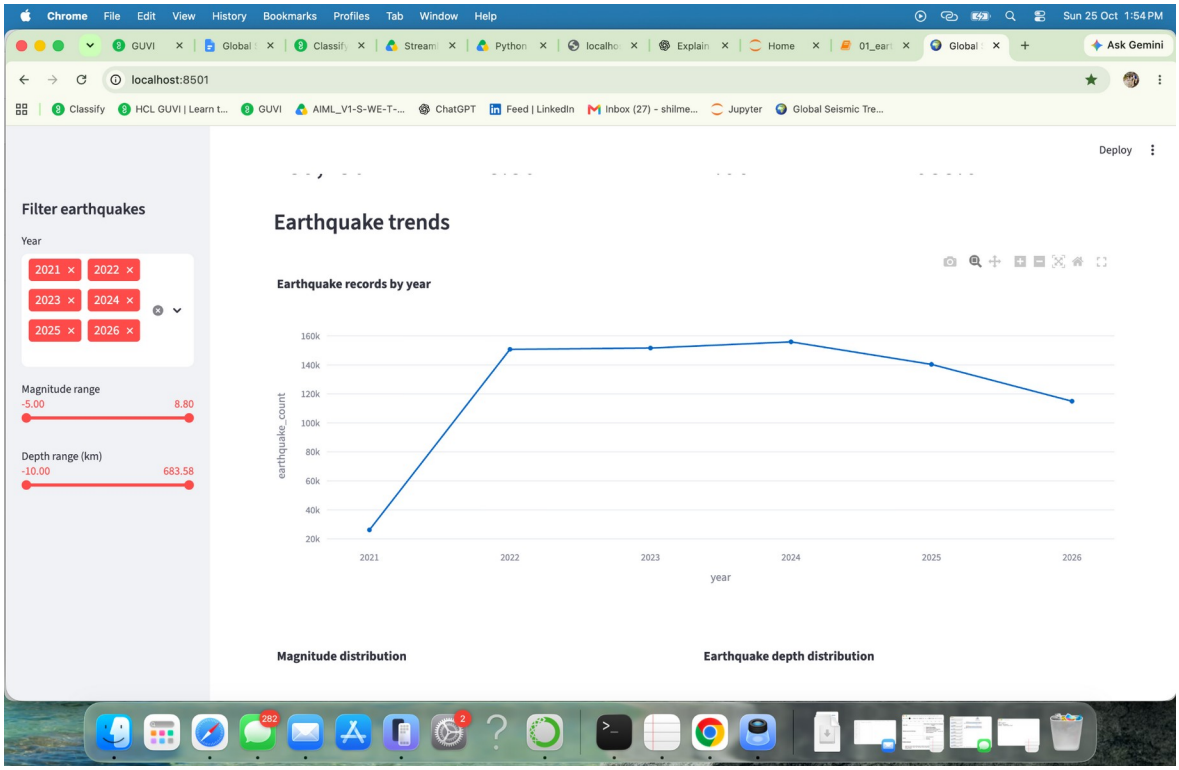


Figure 2. Yearly earthquake counts and magnitude/depth distributions.

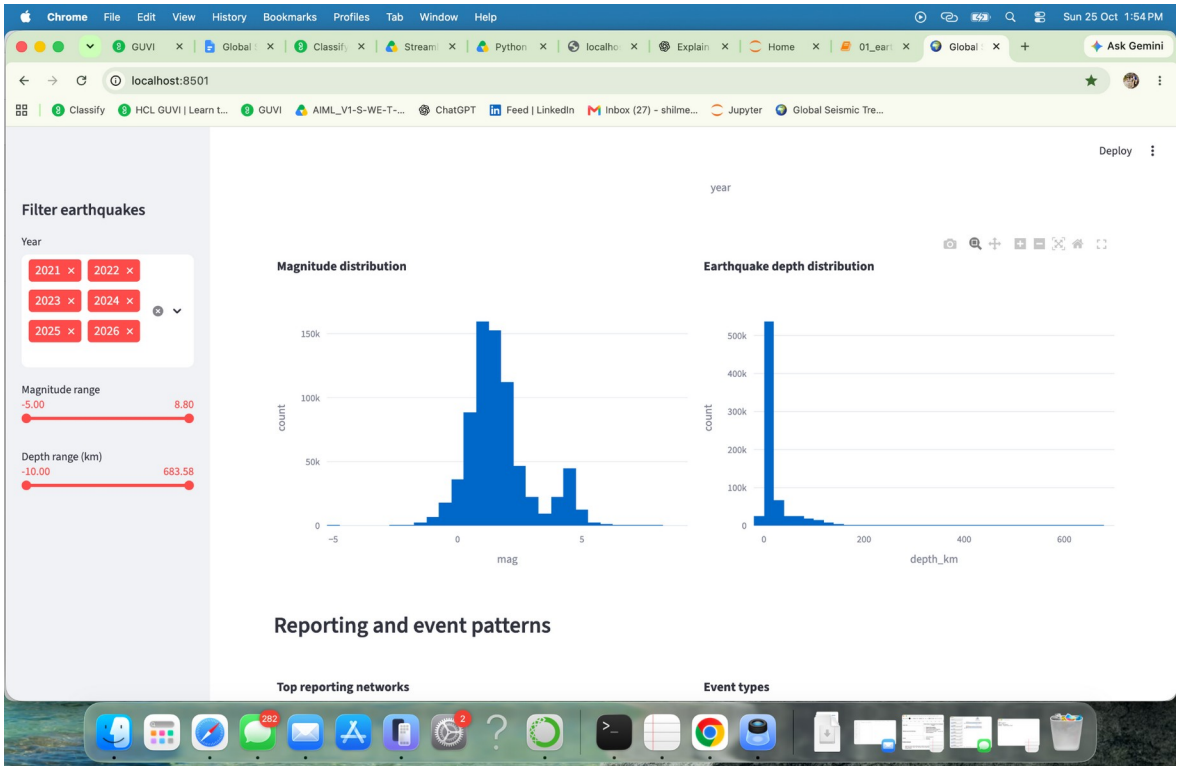


Figure 3. Reporting-network activity and event-type distribution.

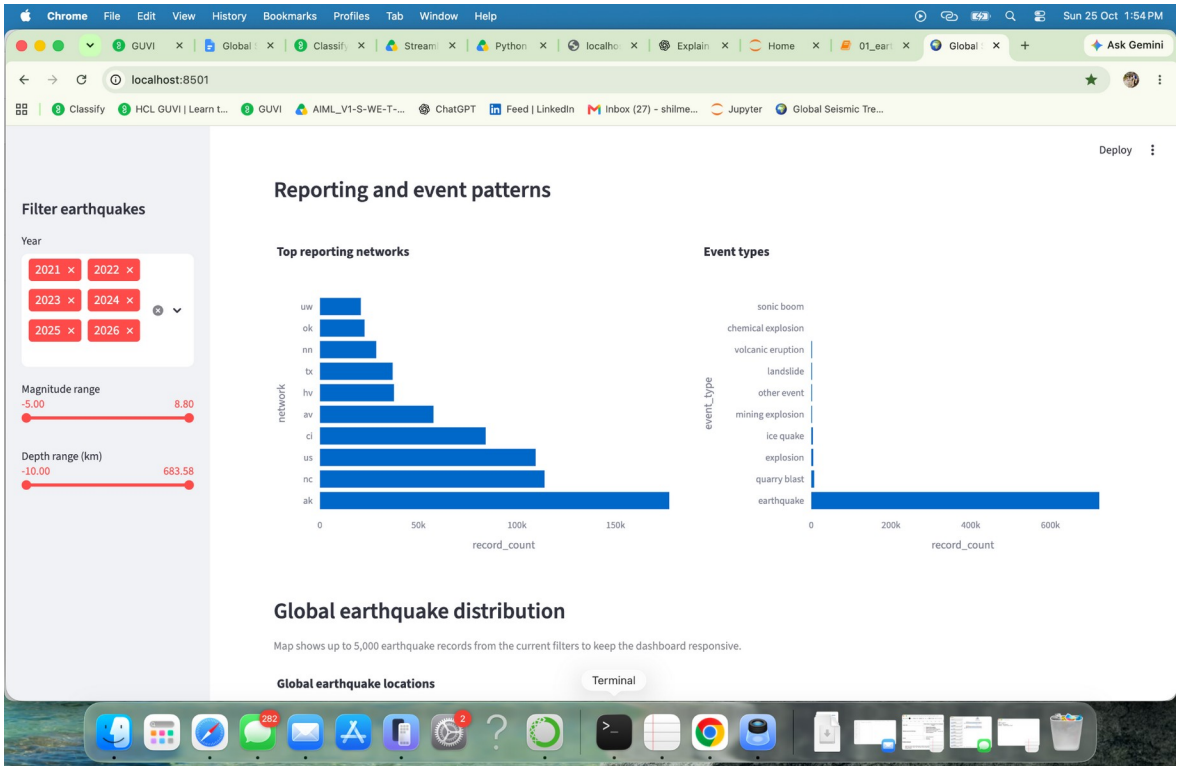


Figure 4. Global earthquake locations plotted on a world map.

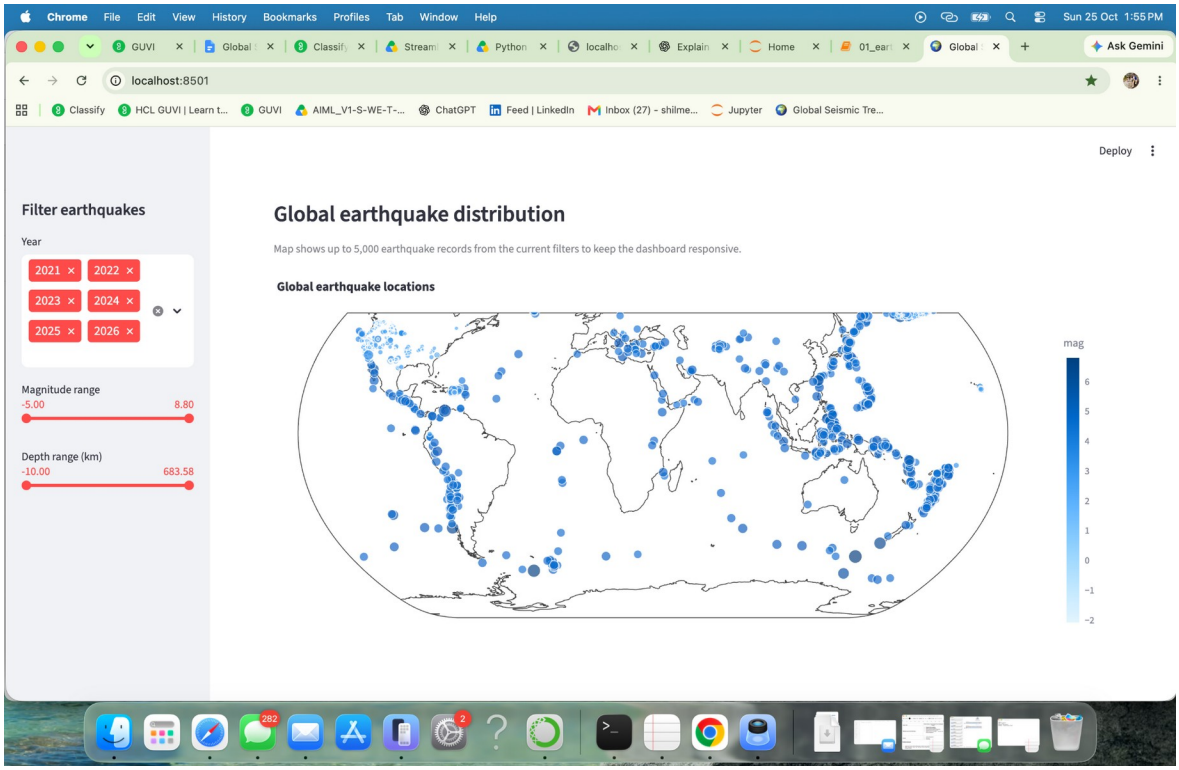


Figure 5. Strongest earthquake records and CSV download option.

10. Limitations and Data-Quality Notes

- The source dataset does not contain casualty counts or estimated economic-loss values in the current schema; questions requiring those fields cannot be answered reliably without an additional source.
- The current schema does not contain an alert-level field; alert-based comparisons cannot be calculated from this dataset alone.
- Country and continent summaries exclude unmatched geographic points and may underrepresent ocean earthquakes.
- Magnitude types are produced by different methods and scales. Their averages are not directly comparable as a measure of severity, especially where sample sizes are very small.
- Counts by reporting network reflect reporting coverage and data availability, not only seismic activity.
- 2021 and 2026 are partial years, so annual totals and growth rates involving those years are not directly comparable with complete years.
- The map intentionally samples up to 5,000 records from the current filter selection for responsiveness; it is not a complete plotted representation of all records.
- Thresholds such as shallow <50 km, deep >=50 km, deep-focus >300 km, or high-gap/high-RMS categories are analysis definitions and should be stated explicitly.

11. How to Run the Dashboard Locally

Prerequisites: Python, Streamlit, Pandas, Plotly, mysql-connector-python, and a running MySQL server.

1. Start the MySQL server and confirm that the global_seismic_trends database and earthquakes table are available.
2. Check the local .streamlit/secrets.toml file and confirm that it contains the database connection settings. Do not include the password in a shared report or public repository.
3. Open Terminal and change directory to the folder containing app.py, for example: cd /Users/shilpathamaraiselvan
4. Run the application with: streamlit run app.py
5. Open the local URL shown by Streamlit in the browser, usually http://localhost:8501.

12. Conclusion

The project demonstrates an end-to-end data workflow combining API-based data retrieval, Python preprocessing, MySQL storage and querying, geographic analysis, and an interactive Streamlit dashboard. It enables users to explore earthquake strength, depth, timing, reporting patterns, event classifications, and global distribution. The documented limitations distinguish conclusions supported by the USGS records from analyses that require additional casualty, economic-loss, or alert-level data.

Appendix A. Suggested Project Folder Structure

Global_Seismic_Trends/

app.py

.streamlit/secrets.toml (local only; do not share)

01_earthquake_data_retrieval.ipynb

earthquakes_5years_processed.csv

analysis_outputs/ (CSV summaries and result files)

README.md

requirements.txt

Global_Seismic_Trends_Project_Report.docx